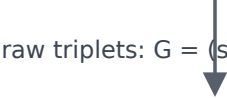


Input — Text Corpus
unstructured documents



1 • Initial Typed Triplet Extraction
Schema-free LLM extraction



raw triplets: $G = (s, t_s, r, o, t_o, Q)$

2 • Relation Deduplication
Embed, cluster, LLM-verify relations



R^* — canonical relation vocabulary

3 • Type Canonicalization
Embed, cluster, LLM-verify types



T^* — canonical type vocabulary

4 • Hierarchy Induction
Priority-ordered, LLM-adjudicated placement



single-parent type hierarchy

5 • Entity Deduplication
Hierarchy-gated, LLM-verified merging



next step -- no data dependency

6 • Constraints Construction
Direction resolution, LCA, PCA scoring



Output — Typed Knowledge Graph
 R^* , T^* +hierarchy, entities, constraints

